TEACHER GUIDE

Cambridge IGCSE Business (0450 / 0264)

Break-even

4 Operations management
4.4 Costs, scale of production and break-even analysis (0450: 4.2)



A revision lesson. Pupils run a car dealership in firms of four and revise break-even on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

1 Cars to break even

Round 1

They take up to 8 cars.

Cars to break even?

Livery and set-up, one-off	**£8,750**
Customer pays per car	**£9,750**
You pay per car	**£8,000**

ANSWER

5 cars

£9,750 − £8,000 = £1,750;
£8,750 ÷ £1,750 = 5

Why: Each car adds £1,750, so 5 cars pay the set-up.

2 Lower break-even

Round 1

You want to lower your break-even point.

Which two changes would lower it?

- **A** Pay more for each car
- **B** Sell more cars
- **C** Pay higher rent
- **D** Raise the selling price
- **E** Cut the fixed costs

ANSWER

✓ Raise the selling price
✓ Cut the fixed costs

Why: More added per car or lower fixed costs; selling more does not move it.

3 Rank by break-even

Round 2

Three deals: fixed costs, and what each car adds (price minus cost).

Order by break-even, fewest cars first?

- **A** £4,000 fixed, £500 a car
- **B** £9,000 fixed, £1,500 a car
- **C** £6,000 fixed, £1,500 a car

ANSWER

1. £6,000 fixed, £1,500 a car
2. £9,000 fixed, £1,500 a car
3. £4,000 fixed, £500 a car

Why: Fixed costs ÷ what each car adds: 4, 6 and 8 cars.

Topic tasks

4 Profit on the deal

Round 2

A deal with a set-up cost.

Profit on the deal?

Livery and set-up, one-off	**£7,000**
Customer pays per car	**£10,750**
You pay per car	**£9,000**
They order	**9 cars**

ANSWER

£8,750

9 × (£10,750 − £9,000) − £7,000 = £8,750

Why: The first cars pay the set-up; the rest is profit: £8,750.

5 Less risky deal

Round 3

Both deals are for 10 cars. The taxi deal breaks even at 4 cars, the hire deal at 8.

Which deal is less risky, and why?

A The taxi deal

B The hire deal

ANSWER

✔ **The taxi deal**

Why: The taxi deal's margin of safety is 6 cars; the hire deal's only 2.

6 New break-even

Round 3

They want £400 off a car.

New break-even in cars? Round up.

Livery and set-up, one-off	**£5,500**
Customer pays per car now	**£19,250**
You pay per car	**£17,000**
Price cut per car	**£400**

ANSWER

3 cars

£18,850 − £17,000 = £1,850; £5,500 ÷ £1,850 → 3

Why: Less added per car (price minus cost) means more cars to break even.

Final questions

Each pupil answers 6 final questions on break-even. They come from these 8 questions.

1 At the break-even level of output, what is true for a dealership?

✔ **Total revenue equals total costs**

2 A dealer's break-even output is 20 cars a month. In May it sells 24 cars. What does this mean?

✔ **It makes a profit in May**

3 A dealer's showroom rent rises. Nothing else changes. What happens to its break-even output?

✔ **It rises**

4 A valeting firm raises its price per valet. Nothing else changes. What happens to its break-even output?

✔ **It falls**

5 On a break-even chart, where is the break-even point?

✔ **Where the total revenue line crosses the total costs line**

6 A dealer's break-even output is 15 cars a month. In March it sells 11 cars. Which statement is correct?

✔ **It makes a loss in March**

7 A cheaper parts supplier cuts the variable cost of each service. Nothing else changes. What happens to break-even output?

✔ **It falls**

8 Why does a new dealer work out its break-even output before opening?

✔ **To know how many cars it must sell to avoid a loss**